

## Point-slope form



- 1 State what form is the equation  $y = 3x + 6$  written in.
- 2 Write down the point-slope form of the equation of a non-vertical line with slope  $m$  that passes through the point  $(x_1, y_1)$ .
- 3 Given a straight line with slope 1 going through point  $A(2, 4)$ , is the slope between  $A$  and any other point on the line always 1?
- 4 For each of the following:
  - i By substituting into the equation  $y = mx + b$ , find the value of  $b$  for this line.
  - ii Hence, write down the equation of the line in slope-intercept form.
  - iii Find the  $x$ -intercept of the line.
  - iv Sketch the graph of the line.
  - a A line has slope  $\frac{2}{3}$  and passes through the point  $(-6, 2)$ .
  - b A line has slope  $-4$  and passes through the point  $(3, -8)$ .
- 5 A line has slope 2 and passes through the point  $(-1, 5)$ .
  - a By substituting into the equation  $y = mx + b$ , find the value of  $b$  for this line.
  - b Hence, write down the equation of the line in slope-intercept form.
  - c Use the point-slope formula to arrive at the same equation from part (b). Show your work.
  - d Sketch the graph of the line.
- 6 A line has a slope of  $-10$  and intersects the  $y$ -axis at  $-1$ .
  - a Use the point-slope formula to find the equation of the line in standard form.
  - b Does the point  $(-7, 68)$  lie on this line?
- 7 Use the point-slope formula to find the equation of the following lines in standard form.
  - a A line passes through the point  $(3, -5)$ , and has a slope of  $-\frac{9}{2}$ .
  - b A line has a slope of  $\frac{5}{9}$  and intersects the  $x$ -axis at  $-10$ .

- 8  $A(-1, 4)$  is a point on a straight line.



If the slope of the line is 4, determine whether the following statements are correct.

a  $\frac{x+1}{y-4} = 4$

b  $\frac{y-4}{x+1} = 4$

c  $\frac{y+1}{x-4} = 4$

d  $\frac{4-y}{-1-x} = 4$

- 9 Each of the following pairs of points define a line:

i Find the slope of the line.

ii Find the equation of the line in slope-intercept form.

a  $(0, 5)$  and  $(5, 25)$

b  $(0, -9)$  and  $(3, 12)$

c  $(0, 3)$  and  $(-7, 31)$

d  $(0, 8)$  and  $(8, 80)$

e  $(0, -6)$  and  $(-10, -56)$

f  $(6, 3)$  and  $(9, 7)$

g  $(1, 3)$  and  $(-3, -13)$

h  $(2, -7)$  and  $(-5, 6)$

- 10 A line passes through the points  $A(3, 4)$  and  $B(-3, -9)$ .

a Find the slope of the line.

b Find the equation of the line using the point-slope formula, and any one point. Express your answer in the form  $Ax + By = C$  with  $A > 0$ .

- 11 Using the point-slope formula, find the equation of the following lines in slope-intercept form.

a A line passes through  $(-2, -9)$  and has a slope of  $-2$ .

b A line passes through  $\left(\frac{7}{8}, -5\right)$  and has a slope of  $2$ .

c A line passes through  $(7, 6)$ , and has a slope of  $\frac{9}{2}$ .

d A line passes through  $(-4, -2)$ , and has a slope of  $3$ .

e A line passes through  $(-6, 1)$  and has a slope of  $4$ .

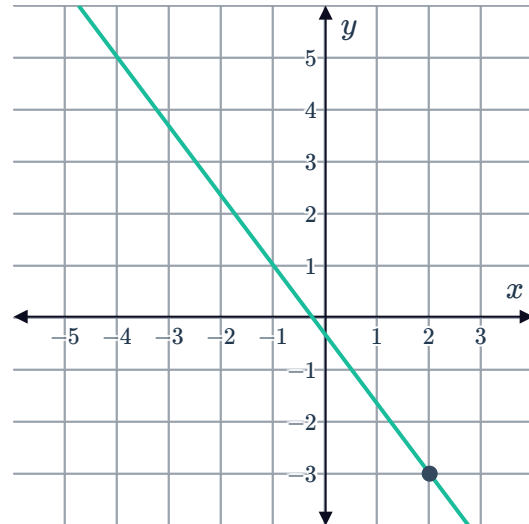
f A line passes through  $\left(1, -\frac{3}{4}\right)$  and has a slope of  $3$ .

g A line passes through  $(-5, 9)$  and has a slope of  $\frac{3}{2}$ .



## Additional questions

- 12 State the equation of the given linear function in point-slope form.



- 13 Sketch the graph of the lines on the coordinate plane given the following equations:

a  $y + 8 = 3(x + 2)$

b  $y + 1 = -2(x + 1)$

c  $y + 6 = -\frac{1}{3}(x - 3)$

d  $y + 2 = \frac{1}{4}x$

- 14 Sally receives an order to make key chains for an upcoming festival. She is told to make at least 50 and at most 100 key chains, and will be paid \$8 for each key chain she makes. Sally makes 80 key chains and is paid \$665 altogether.
- a Let  $x$  represent the number of key chains Sally makes and  $y$  represent the amount she is paid. Write an equation in point-slope form relating  $x$  and  $y$ .
  - b Convert the equation from part (a) to slope-intercept form.
  - c Determine the flat fee that Sally was paid in addition to the \$8 per key chain that she received.
  - d State the domain of this linear relationship



- 15 Gary is flying in a plane which is moving directly away from his home at a constant speed. At 2 hours into the flight, he checked that he was 1042 miles from his home. He checks again at 7 hours into the flight to find that he is now 3542 miles from his home.
- Find the speed of the plane, in miles per hour.
  - Write an equation in point-slope form relating the number of hours flying,  $x$ , to the distance Gary is from his home,  $y$ , in miles.
  - If the destination of Gary's flight is 5542 miles from his home, find its duration.
- 16 Create a scenario where you would choose to use each of the following forms and explain why you would use that form over the others:
- Point-slope form,  $(y - y_1) = m(x - x_1)$
  - Slope-intercept form,  $y = mx + b$
  - Standard form,  $Ax + By = C$
- 17 Rewrite the point-slope form of a linear equation in slope-intercept form. Draw connections between the two forms.
- 18 A cup is placed under a faucet which is leaking at a rate of 2 oz per hour. After 4 hours, the cup contains 15 oz of water.
- Write an equation in point-slope form relating the volume in the cup,  $y$ , to the amount of time that has passed in hours,  $x$ .
  - Now, rewrite the equation into slope-intercept form.
  - State and interpret the  $y$ -intercept of the linear equation.
  - Suppose that the cup has a maximum capacity of 18 oz. Determine how long it can be under the faucet before it starts to over flow. Now, state the domain and range.
- 19 Paige is planning an event and the brochure for the caterer at the venue shows the following graph for the meal costs.
- State and interpret the domain of the linear function.
  - Paige concludes that the unit cost per person is \$16.25. Determine if this is correct. If so, explain how to calculate this value. If not, explain the error and state the correct value.
  - State the equation of the linear function in point-slope form

